E350

U1300 and U3400

RELEASE NOTES FOR F/W VERSION 8.0

Date: 25/02/2021
Filename: E350 U1300 and U3400 Release Notes
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Revision history

Version	Date	Comments
1	22/02/2021	Initial Release
2	25/02/2021	Correct U3400 build number.

Important Information
This document provides information relating to E350 U1300 and U3400 meter firmware (the “Product”)

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Table of contents

1 About this document 4
1.1 Purpose & Scope 4
1.2 Definitions, Acronyms, and Abbreviations 4
1.3 References 4
2 About the release, compatible products, and Summary 5
2.1 New Features for Supported Meters 6
2.1.1 General New Features 6
2.1.2 Firmware Upgrade 6
2.1.3 Firmware Downgrade 6
2.1.4 Ignore Load Control timetable delay on power-up 7
2.1.5 Tamper Event Grace Period 7
2.1.6 Smart Time Synchronisation improvement when using SSN/Itron Modules 7
2.1.7 Ripple Failsafe Event enhancement. (TFS-1014925) 7
2.1.8 Firmware support to drive the new U3400 S3 Supply Control Switches. 8
2.1.9 Energy Redirection when the Boost button is pressed on a U3400 meter. 8
2.2 Command Center 8
2.2.1 Command Center Software Quality Assurance testing 8
2.3 Bug fixes in this release 8
2.4 Known Issues in this release 9

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1 About this document

1.1 Purpose & Scope

The purpose of this ***8.0 Firmware Release Note*** is to communicate new features, limitations, and changes in meter functionality.

The 8.0 firmware release is for the E350 - U1300 and U3400 meters.

1.2 Definitions, Acronyms, and Abbreviations

RSM:	Relay State Machine for the load control relay
LC:	Load Control
LP:	Load Profile
CLC/R:	Controlled Load Contactor or Relay
CSM:	Customer Supply Monitoring including Neutral Integrity
QoS:	Quality of Supply
AMI:	Advanced Metering Infrastructure
AD:	Auto Disconnect
S2:	Series 2
SCC:	Supply Capacity Control
STS:	Smart Time Synchronisation
HES:	Head End Systems
E350:	Family name of Landis+Gyr modular meter including the types U1200, U1300, U1300 S2 U3300, U3400 and U3350
E650:	Family name of Landis+Gyr commercial and industrial meter including the type EM5400 and EM5400 S2
E420:	Family name of Landis+Gyr integrated meter S1100 RF and S1100 LTE
R1100:	Landis+Gyr’s smart load control device.

1.3 References

E350 U1300 User Manual

E350 U3400 User Manual

EMPWin+ User Manual

2 About the release, compatible products, and Summary

This firmware package provides upgrade to the following meter/firmware:

Previously Released firmware version: 7.0

This Released Version: 8.0

Build Version: 013 (U3400) and 007 (U1300)

U1300		U3400	
Firmware Upgrade*	**Firmware Downgrade***	**Firmware Upgrade***	**Firmware Downgrade***
4.2 → 5.4 (Series 1 Only) ↓ 4.3 → 5.4 (Series 1 Only) ↓ 4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.8 → 8.0 ↓ 4.9 → 8.0 ↓ 5.0 → 8.0 ↓ 5.4 → 8.0 ↓ 6.0 → 8.0 ↓ 7.0 → 8.0	8.0 ↓ 7.0 ↓ 6.0	4.4 → 5.4 ↓ 4.5 → 5.4 ↓ 4.6 → 5.4 ↓ 4.7 → 5.4 ↓ 4.9 → 8.0 ↓ 5.0 → 8.0 ↓ 5.4 → 8.0 ↓ 6.0 → 8.0 ↓ 7.0 → 8.0	8.0 ↓ 7.0 ↓ 6.0

Important Note: *customer will be required to refer to the release notes for the particular firmware version to understand the limitations of the upgrade/downgrade. Please also refer to the intermediary firmware release notes when skipping meter firmware version.

This firmware has been tested on the following platforms:

- E350 U1300 meters (this includes Series 1, Series 2, Series 3, 16Mb and 64Mb Variants)
- E350 U3400 meters (this includes Series 1, Series 3, 16Mb and 64Mb Variants)
- EMPWin+ 4.0.38

The FT100 release 5.5 or earlier must not be used with 8.0 meter firmware.

EMPWin+ 4.0.38 or greater must be used with 8.0 meter firmware.

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2.1 New Features for Supported Meters

2.1.1 General New Features

The E350 U1300 & U3400 meters with 8.0 firmware now have the following enhancements.

- Ignore random delay for LC timetables on power-up.
- Tamper Event Grace Period
- Smart Time Synchronisation improvement when using SSN/Itron Modules.
- Ripple Failsafe Event enhancement.
- Firmware support to drive the new U3400 S3 Supply Control Switches.
- Energy Redirection when the Boost button is pressed on a U3400 to control the Load Control relay.

2.1.2 Firmware Upgrade

- The E350 U1300 & U3400 meters with 16Mb or 64Mb memory can be upgraded to 8.0 Firmware. However, there will be a Non Volatile error 0x08 after F/W upgrade from Rx.y to 06.00, 07.00 or 08.00 this is due to the significant changes to the LP configuration and increased LP memory size.
- There for when upgrading the firmware, you will be required to perform a full reprogramming of the meter.
- U3400 S3 meters will be manufactured with 8.0 firmware or greater.

Firmware Upgrade Path	Full Reprogram	Rewrite Meter Passwords
R4.9 and earlier upgrade to 8.0	Yes	No
R5.0 and R5.4 upgrade to 8.0	Yes	No
6.0 upgrade to 8.0	Yes	No
7.0 upgrade to 8.0	Yes	No

2.1.3 Firmware Downgrade

Downgrade back from 8.0 to 7.0 is supported but any features that are unique to 8.0 will become disabled after the downgrade. Downgrade to any earlier version than 6.0 is not supported. Doing so might require the meter being to be returned to Landis & Gyr to be factory reprogrammed with the F/W version and program that is required. There is no automatic protection in the firmware that would prevent a customer from performing this type of data destructive operation.

2.1.4 Ignore Load Control timetable delay on power-up

When this feature is enabled in the meter program or ripple program and a power outage occurs while there is a pending delayed LC operation (Timetable Random Delay), then the Timetable random delay will be ignored and just the “Power Up Restore Delay” will be applied. After this all-other LC Delayed (Timetable Delays) operations will apply. New Table 2132 was added to support this feature and F/W automatically enables the correct access permissions. This feature is applicable to all LC relays present in Meter.

The Ignore random delay selection is available in the Meter and the Ripple program. The Ignore Load Control timetable delay on power-up feature in the ripple program will be used when ripple is in control of the relays.

2.1.5 Tamper Event Grace Period (Licenced Feature)

If this Tamper Event Grace Period is enabled, then after full programming a grace period will be observed. Therefore any terminal cover tamper events will be suppressed for the user selected time of 0 to 60 minutes (0 = no Grace period). If the meter is power cycled during the grace period, then the grace period will be suspended during the power down time and will resume on power up. This is accurate to the nearest minute. Table 2131 was added to store the internally used count down value. Tamper Event Grace Period is a licenced feature. Please contact L+G if you are interested in enabling this feature.

2.1.6 Smart Time Synchronisation improvement when using SSN/Itron Modules.

The Smart Time Synchronisation feature will no longer check if an SSN comms module is connected before allowing the Smart Time Synchronisation (STS) feature to be used to slowly adjust a meter’s time backwards without stopping Load Profile (by crossing back into a previously completed LP interval). Earlier F/W versions would block the STS method if an SSN module was detected because SSN had their own time adjustment logic.

2.1.7 Ripple Failsafe Event enhancement. (TFS-1014925)

Ripple Failsafe Event 2120 has been enhanced to report which Load Control relay triggered the exit of the event.

Events

Time	Description
08/07/2020 14:08:28	Ripple event. Ripple Failsafe Exit. Recovery: Auto LC2
08/07/2020 14:08:28	Ripple event. Ripple Failsafe Exit. Recovery: Auto LC1
08/07/2020 14:08:19	Ripple event. Ripple Failsafe Exit. Recovery: Auto LC2
08/07/2020 14:08:19	Ripple event. Ripple Failsafe Exit. Recovery: Auto LC1

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2.1.8 Firmware support to drive the new U3400 S3 Supply Control Switches.

The U3400 Series 3 meter includes 3 individual Supply Control Switches.
The U3400 Firmware has been enhanced to drive all three Supply Control Switches together in one operation. To ensure there is sufficient capacity in the meters power supply under various network conditions the meter will operate the individual SCS 300mS apart.
The U3400 S3 meter, Breaker open and close operations can be performed faster than in previous U3400 S1 hardware versions.

2.1.9 Energy Redirection when the Boost button is pressed on a U3400 meter.

The U3400 now allows the user to select which phase the energy will be recorded to when the Boost button is pressed to close the LC relay. In previous F/W versions the energy was recorded in Phase C because that is where the Load Control relay is connected. MeterDev for the U3400 has three choices Phase A or B or C. Phase C is the default for all earlier F/W. Should a meter program with Phase A or B, be selected and written into a meter with an earlier firmware version. The meter will continue to record the energy in Phase C.

2.2 Command Center

2.2.1 Command Center Software Quality Assurance testing.

Command Center (CC) version 7.6.0.23 has been tested with 8.0 F/W.

2.3 Bug fixes in this release

Item No.	*Description*
TFS-857014	Missing Event log 2118 (Extended Load Control) for Four Season LC Change in Ripple Time Table. Ripple Timetable season change events are now logged as they occur at midnight.
TFS-1052780	In all previous F/W versions if the test LED was programmed to flash for VARh energy and then the optical port was used to perform long reads then the VARh LED would turn on and stay on making the counting of VARH energy via the LED impossible. If the LED were used for Wh testing it would pulse correctly regardless of the optical port usage. The bug has now been fixed in Firmware 8.0.
TFS-857051 **TFS-48970** **TFS-1565** **Mantis-3918**	With previous firmware versions the U1300 meters, LC1 & LC2 random delay closed values tend to have the same minutes and seconds with only the hours being different. Random delay was not random enough for the two relays. The bug has now been fixed in Firmware 8.0.
TFS-1052780	It was discovered during testing the VARH LED could change state during optical communications. This did not affect metrology or the integrity of the billing or load profile data in the meter. The bug has now been fixed in Firmware 8.0.
TFS-888375	F/W 07.00.006 did not automatically enable R/W access to new table 2130. When upgrading to F/W 07.00.006 the password permissions were required to be written to the meter after the firmware upgrade. The bug has now been fixed in Firmware 8.0 and it is no longer a requirement to write password permissions to the meter when upgrading to Firmware 8.0.

2.4 Known Issues in this release.

Item No.	*Description*
TFS-1066340	It has been noticed during Transparent Mode testing with latest L+G E35C comms module and F/W 14.69, the LP reads can fail if the amount of data that needs to be read is large (more than 60 days but varies). New comms F/W 14.70 has been created to improve the reading reliability of Transparent mode reads when using MV90 and large amount of LP data.
TFS-48967 **TFS-857816** **Mantis-3906**	Demand Limit Load Control If Under Frequency Load Control (UFLC) module is triggered due to bad frequency, then the frequency recovers and UFLC starts its Nominal Delay period AND then a short power cycle occurs (about 3 seconds) AND on Power Up the Frequency is still within Normal limits; under these conditions the UFLC module might not correctly exit its Nominal waiting period. This can cause the UFLC module to block other lower priority RSM commands such as Ripple telegrams. It appears that when this incorrect state is reached that it might require two power cycles to correctly reset the RSM logic.
TFS-857835 **TFS-1566** **TFS-48701** **Mantis-3919**	It was noticed the Ripple Control (RC) icon on the LCD may not display correctly under the following scenario. The RC icon on the display should be slowly flashing if the meter contains a valid ripple program and it is in Failsafe mode. Writing meter program with Ripple support "False" settings to meter while it is in Failsafe mode should automatically cancel any ripple operation and set RC icon to permanent On state. It was noticed the RC icon was not set to permanent On state and continued flashing slowly.

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